Tutorial- Week 4

RegNo.: RA2511033010078

Name: V.I. Sahaana

Course: CSE Software Engineering

Section: AM2

1)

```java
import java.util.Scanner;

public class Product {  3 usages
    int productid;  2 usages
    String productname;  2 usages
    float price;  3 usages
    int qty;  3 usages
    Product(int id, String name, float p, int q){  1 usage
        productid=id;
        productname=name;
        price=p;
        qty=q;
    }
    void display(){  1 usage
        double total=price*qty;
        if(total>=5000){
            total = total-(0.1*total);

        }
        else{
            total = total-(0.05*total);
        }
        System.out.println("ID:"+productid);

        System.out.println("Name:"+productname);
        System.out.println("Price:"+price);
        System.out.println("Quantity:"+qty);
        System.out.println("Final Price:"+total);
        System.out.println("\n");
    }
}
```

```java
}
class ProductDetails{
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        Product[] products = new Product[5];
        for (int i=0;i<5;i++) {
            System.out.println("Enter Details of Product"+(i+1)+":");
            System.out.println("Enter ProductID:");
            int id = sc.nextInt();
            sc.nextLine();
            System.out.println("Enter name:");
            String name = sc.nextLine();
            System.out.println("Enter price:");
            float p= sc.nextFloat();
            sc.nextLine();
            System.out.println("Enter Quantity:");
            int q=sc.nextInt();
            sc.nextLine();
            System.out.println('\n');
            products[i]=new Product(id,name,p,q);
        }
        for (int i=0;i<5;i++){
            products[i].display();
        }
    }
}
```

Output:

```
C:\Users\Sahaana\.jdks\openjdk-14.0.2\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Co
Enter Details of Product1:
Enter ProductID:
101
Enter name:
Laptop
Enter price:
50000
Enter Quantity:
1


Enter Details of Product2:
Enter ProductID:
102
Enter name:
PC
Enter price:
12000
Enter Quantity:
1


Enter Details of Product3:
Enter ProductID:
103
Enter name:
TV
```

```
Enter ProductID:
103
Enter name:
TV
Enter price:
60000
Enter Quantity:
1


Enter Details of Product4:
Enter ProductID:
104
Enter name:
AC
Enter price:
50000
Enter Quantity:
3


Enter Details of Product5:
Enter ProductID:
105
Enter name:
Dining Table
Enter price:
40000
Enter Quantity:
```

```
Enter Quantity:
1


ID:101
Name:Laptop
Price:50000.0
Quantity:1
Final Price:45000.0


ID:102
Name:PC
Price:12000.0
Quantity:1
Final Price:10800.0


ID:103
Name:TV
Price:60000.0
Quantity:1
Final Price:54000.0


ID:104
Name:AC
Price:50000.0
```

```
ID:104
Name:AC
Price:50000.0
Quantity:3
Final Price:135000.0


ID:105
Name:Dining Table
Price:40000.0
Quantity:1
Final Price:36000.0



Process finished with exit code 0
```

2)

```java
import java.util.Scanner;

class Student{  4 usages
    int roll;  3 usages
    String name;  3 usages
    float m1,m2,m3;  3 usages
    double attendance,av;  3 usages
    float total;  4 usages
    Student(int roll, String name, float m1, float m2, float m3, double attendance){  1 usage
        this.roll=roll;
        this.name=name;
        this.m1=m1;
        this.m2=m2;
        this.m3=m3;
        this.attendance=attendance;

    }
    void calculate(){  1 usage
        total=m1+m2+m3;
        System.out.println("Total="+total);

    }
    void average() {  1 usage
         av=total/3;
        System.out.println("Average=" + av);
    }

    }

```

```java
class StudentPerformance{
    public static void main(String[] args) {
        Student[]students=new Student[5];
        Scanner sc= new Scanner(System.in);
        for(int i=0;i<5;i++){

            System.out.println("Enter Student Details\n");
            System.out.println("Enter details for Student "+(i+1));
            System.out.println("Enter Roll:");
            int roll=sc.nextInt();
            sc.nextLine();
            System.out.println("Enter Name:");
            String name=sc.nextLine();
            System.out.println("Enter Marks for 3 subjects:");
            float m1= sc.nextFloat();
            float m2= sc.nextFloat();
            float m3= sc.nextFloat();
            System.out.println("Enter Attendance Percentage:");
            double attendance= sc.nextDouble();
            students[i]=new Student(roll, name, m1, m2, m3, attendance);
            students[i].calculate();
            students[i].average();
            System.out.println("\n");

        }
        for (int i = 0; i < 5; i++) {

            System.out.println("\n----- Student Details -----");
            System.out.println("Roll Number : " + students[i].roll);
            System.out.println("Name        : " + students[i].name);
```

```
            System.out.println("Name        : " + students[i].name);
            System.out.println("Mark 1      : " + students[i].m1);
            System.out.println("Mark 2      : " + students[i].m2);
            System.out.println("Mark 3      : " + students[i].m3);
            System.out.println("Total       : " + students[i].total);
            System.out.println("Average     : " + students[i].av);
            System.out.println("Attendance  : " + students[i].attendance + "%");

            if (students[i].av >= 50)
                System.out.println("Result      : Pass");
            else
                System.out.println("Result      : Fail");

            if (students[i].av >= 75 && students[i].attendance >= 80)
                System.out.println("Scholarship : Eligible");
            else
                System.out.println("Scholarship : Not Eligible");

            if (students[i].av >= 85)
                System.out.println("Performance : Excellent");
            else
                System.out.println("Performance : Good");
        }
        Student topStudent = students[0];

        for (int i = 1; i < 5; i++) {
            if (students[i].av > topStudent.av) {
                topStudent = students[i];
```

```
                topStudent = students[i];
            }
        }

        System.out.println("\n----- Student with Highest Average -----");
        System.out.println("Roll Number : " + topStudent.roll);
        System.out.println("Name        : " + topStudent.name);
        System.out.println("Average     : " + topStudent.av);



    }
}
```

Output:

```
Enter Student Details

Enter details for Student 1
Enter Roll:
1234
Enter Name:
abcd
Enter Marks for 3 subjects:
89
86
87
Enter Attendance Percentage:
75
Total=262.0
Average=87.33333587646484


Enter Student Details

Enter details for Student 2
Enter Roll:
98
Enter Name:
ram
Enter Marks for 3 subjects:
89
96
99
```

```
96
99
Enter Attendance Percentage:
90
Total=284.0
Average=94.66666412353516


Enter Student Details

Enter details for Student 3
Enter Roll:
54
Enter Name:
arun
Enter Marks for 3 subjects:
89
99
96
Enter Attendance Percentage:
89
Total=284.0
Average=94.66666412353516


Enter Student Details

Enter details for Student 4
Enter Roll:
```

```
Enter details for Student 4
Enter Roll:
76
Enter Name:
priya
Enter Marks for 3 subjects:
89
98
99
Enter Attendance Percentage:
78
Total=286.0
Average=95.33333587646484


Enter Student Details

Enter details for Student 5
Enter Roll:
34
Enter Name:
jay
Enter Marks for 3 subjects:
99
97
96
Enter Attendance Percentage:
89
Total=292.0
```

```
89
Total=292.0
Average=97.33333587646484



----- Student Details -----
Roll Number : 1234
Name        : abcd
Mark 1      : 89.0
Mark 2      : 86.0
Mark 3      : 87.0
Total       : 262.0
Average     : 87.33333587646484
Attendance  : 75.0%
Result      : Pass
Scholarship : Not Eligible
Performance : Excellent

----- Student Details -----
Roll Number : 98
Name        : ram
Mark 1      : 89.0
Mark 2      : 96.0
Mark 3      : 99.0
Total       : 284.0
Average     : 94.66666412353516
Attendance  : 90.0%
Result      : Pass
```

```
Attendance  : 90.0%
Result      : Pass
Scholarship : Eligible
Performance : Excellent

----- Student Details -----
Roll Number : 54
Name        : arun
Mark 1      : 89.0
Mark 2      : 99.0
Mark 3      : 96.0
Total       : 284.0
Average     : 94.66666412353516
Attendance  : 89.0%
Result      : Pass
Scholarship : Eligible
Performance : Excellent

----- Student Details -----
Roll Number : 76
Name        : priya
Mark 1      : 89.0
Mark 2      : 98.0
Mark 3      : 99.0
Total       : 286.0
Average     : 95.33333587646484
Attendance  : 78.0%
Result      : Pass
Scholarship : Not Eligible
```

```
Average     : 95.33333587646484
Attendance  : 78.0%
Result      : Pass
Scholarship : Not Eligible
Performance : Excellent

----- Student Details -----
Roll Number : 34
Name        : jay
Mark 1      : 99.0
Mark 2      : 97.0
Mark 3      : 96.0
Total       : 292.0
Average     : 97.33333587646484
Attendance  : 89.0%
Result      : Pass
Scholarship : Eligible
Performance : Excellent

----- Student with Highest Average -----
Roll Number : 34
Name        : jay
Average     : 97.33333587646484

Process finished with exit code 0
```

3)

```java
import java.util.Scanner;

class Employee {  3 usages
    int id;  2 usages
    String name;  2 usages
    double monthlySalary;  5 usages
    double annualSalary;  3 usages
    double bonus;  3 usages
    boolean eligible;  3 usages

    Employee(int id, String name, double monthlySalary) {  1 usage
        this.id = id;
        this.name = name;
        this.monthlySalary = monthlySalary;
    }

    void calculateAnnualSalary() {  1 usage
        annualSalary = monthlySalary * 12;
    }

    void calculateBonus() {  1 usage
        if (monthlySalary >= 30000)
            bonus = annualSalary * 0.10;
        else
            bonus = 0;
    }

    void checkEligibility() {  1 usage
        if (monthlySalary >= 30000)
            eligible = true;
```

```java
    void checkEligibility() {  1 usage
        if (monthlySalary >= 30000)
            eligible = true;
        else
            eligible = false;
    }

    void display() {  1 usage
        System.out.println("\n----- Employee Details -----");
        System.out.println("Employee ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Monthly Salary: " + monthlySalary);
        System.out.println("Annual Salary: " + annualSalary);
        System.out.println("Bonus: " + bonus);

        if (eligible)
            System.out.println("Eligibility: Eligible");
        else
            System.out.println("Eligibility: Not Eligible");
    }
}
class EmployeePerformance {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Employee[] employees = new Employee[5];
        for (int i = 0; i < 5; i++) {
            System.out.println("\nEnter details for Employee " + (i + 1));
            System.out.print("Enter Employee ID: ");
            int id = sc.nextInt();
            sc.nextLine();
```

```
            sc.nextLine();
            System.out.print("Enter Name: ");
            String name = sc.nextLine();
            System.out.print("Enter Monthly Salary: ");
            double monthlySalary = sc.nextDouble();
            employees[i] = new Employee(id, name, monthlySalary);
            employees[i].calculateAnnualSalary();
            employees[i].calculateBonus();
            employees[i].checkEligibility();
        }
        System.out.println("\n      EMPLOYEE DETAILS        ");
        for (int i = 0; i < 5; i++) {
            employees[i].display();
        }
        sc.close();
    }
}
```

Output:

```
Enter details for Employee 1
Enter Employee ID: 1
Enter Name: rahul
Enter Monthly Salary: 100000

Enter details for Employee 2
Enter Employee ID: 2
Enter Name: arush
Enter Monthly Salary: 150000

Enter details for Employee 3
Enter Employee ID: 3
Enter Name: arya
Enter Monthly Salary: 130000

Enter details for Employee 4
Enter Employee ID: 4
Enter Name: keerthi
Enter Monthly Salary: 170000

Enter details for Employee 5
Enter Employee ID: 5
Enter Name: kavya
Enter Monthly Salary: 140000
```

```
      EMPLOYEE DETAILS

----- Employee Details -----
Employee ID: 1
Name: rahul
Monthly Salary: 100000.0
Annual Salary: 1200000.0
Bonus: 120000.0
Eligibility: Eligible

----- Employee Details -----
Employee ID: 2
Name: arush
Monthly Salary: 150000.0
Annual Salary: 1800000.0
Bonus: 180000.0
Eligibility: Eligible

----- Employee Details -----
Employee ID: 3
Name: arya
Monthly Salary: 130000.0
Annual Salary: 1560000.0
Bonus: 156000.0
Eligibility: Eligible

----- Employee Details -----
Employee ID: 4
Name: keerthi
```

```
----- Employee Details -----
Employee ID: 4
Name: keerthi
Monthly Salary: 170000.0
Annual Salary: 2040000.0
Bonus: 204000.0
Eligibility: Eligible

----- Employee Details -----
Employee ID: 5
Name: kavya
Monthly Salary: 140000.0
Annual Salary: 1680000.0
Bonus: 168000.0
Eligibility: Eligible

Process finished with exit code 0
```

4)

```java
import java.util.Scanner;

class Patient {  3 usages
    String name;  2 usages
    double consultationFee;  2 usages
    double discount;  4 usages
    double finalAmount;  2 usages

    Patient(String name, double consultationFee) {  1 usage
        this.name = name;
        this.consultationFee = consultationFee;
    }

    void calculateFinalAmount(double fee) {  1 usage
        if (fee >= 2000)
            discount = fee * 0.10;
        else
            discount = fee * 0.05;

        finalAmount = fee - discount;
    }

    void display() {  1 usage
        System.out.println("\n    Patient Details       ");
        System.out.println("Patient Name: " + name);
        System.out.println("Original Consultation Fee: " + consultationFee);
        System.out.println("Discount: " + discount);
        System.out.println("Final Amount: " + finalAmount);
    }
}
```

```java
class HospitalBilling {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Patient[] patients = new Patient[5];

        for (int i = 0; i < 5; i++) {

            System.out.println("\nEnter details for Patient " + (i + 1));

            System.out.print("Enter Patient Name: ");
            String name = sc.nextLine();

            System.out.print("Enter Consultation Fee: ");
            double fee = sc.nextDouble();
            sc.nextLine();

            patients[i] = new Patient(name, fee);

            patients[i].calculateFinalAmount(fee);
        }

        System.out.println("\n       CONSULTATION BILLING        ");

        for (int i = 0; i < 5; i++) {
            patients[i].display();
        }
    }
}
```

Output:

```
Enter details for Patient 1
Enter Patient Name: rahul
Enter Consultation Fee: 500

Enter details for Patient 2
Enter Patient Name: priya
Enter Consultation Fee: 700

Enter details for Patient 3
Enter Patient Name: ravi
Enter Consultation Fee: 850

Enter details for Patient 4
Enter Patient Name: vijay
Enter Consultation Fee: 560

Enter details for Patient 5
Enter Patient Name: teju
Enter Consultation Fee: 890
```

```
      CONSULTATION BILLING

    Patient Details
Patient Name: rahul
Original Consultation Fee: 500.0
Discount: 25.0
Final Amount: 475.0

    Patient Details
Patient Name: priya
Original Consultation Fee: 700.0
Discount: 35.0
Final Amount: 665.0

    Patient Details
Patient Name: ravi
Original Consultation Fee: 850.0
Discount: 42.5
Final Amount: 807.5

    Patient Details
Patient Name: vijay
Original Consultation Fee: 560.0
Discount: 28.0
Final Amount: 532.0
```

```
    Patient Details
Patient Name: teju
Original Consultation Fee: 890.0
Discount: 44.5
Final Amount: 845.5

Process finished with exit code 0
```

5)

```java
class Ride {  7 usages
    int rideNo;  2 usages
    String rideName;  2 usages

    Ride(int rideNo, String rideName) {  5 usages
        this.rideNo = rideNo;
        this.rideName = rideName;
    }
}

class AmusementPark {
    public static void main(String[] args) {

        Ride[] rides = new Ride[5];

        rides[0] = new Ride( rideNo: 1,  rideName: "Roller Coaster");
        rides[1] = new Ride( rideNo: 2,  rideName: "Ferris Wheel");
        rides[2] = new Ride( rideNo: 3,  rideName: "Bumper Cars");
        rides[3] = new Ride( rideNo: 4,  rideName: "Water Ride");
        rides[4] = new Ride( rideNo: 5,  rideName: "Haunted House");

```

```java
        for (int i = 0; i < 5; i++) {

            System.out.print(rides[i].rideName + " : ");

            for (int j = 1; j <= rides[i].rideNo; j++) {
                System.out.print("*");
            }

            System.out.println();
        }
    }
}
```

Output:

```
C:\Users\Sahaana\.jdks\openjdk-14.0.2\bin\java.exe "-javaagent:C:\Program Files\JetBrains\
Roller Coaster : *
Ferris Wheel : **
Bumper Cars : ***
Water Ride : ****
Haunted House : *****

Process finished with exit code 0
```